

Session 5: Joins and Relationships Trainer Coverage: Foreign keys and primary keys. Inner Join, Left Join, Right Join. Demo: Link users with their transactions table. Pending

Tasks

- 1) Create two tables in SQL: influencers (with influencer_id as PRIMARY KEY and name) and posts (with post_id as PRIMARY KEY, influencer_id as FOREIGN KEY, and caption). Insert at least 3 influencers and 2 posts for each influencer.

Query:

```
1 CREATE TABLE influencers (  
2     influencer_id INT PRIMARY KEY AUTO_INCREMENT,  
3     name VARCHAR(100) NOT NULL  
4 );  
5  
6 CREATE TABLE posts (  
7     post_id INT PRIMARY KEY AUTO_INCREMENT,  
8     influencer_id INT,  
9     caption VARCHAR(255),  
10    FOREIGN KEY (influencer_id) REFERENCES influencers(influencer_id)  
11 );  
12  
13 INSERT INTO influencers (name) VALUES  
14 ('abhi'),  
15 ('velcy'),
```

```
16 ('harish');  
17  
18 INSERT INTO posts (influencer_id, caption) VALUES  
19 (1, 'My first travel vlog!'),  
20 (1, 'Exploring a new place today!'),  
21  
22 (2, 'Morning workout routine!'),  
23 (2, 'Healthy food ideas!'),  
24  
25 (3, 'New technology tips!'),  
26 (3, 'Learning SQL today!');
```

Run SQL query/queries on database my database: 

```
1 SELECT * FROM influencers;
2
3 SELECT * FROM posts;
```

n SQL query/queries on table my database.posts: 

```
1 SELECT
2     influencers.name,
3     posts.caption
4 FROM influencers
5 JOIN posts
6 ON influencers.influencer_id = posts.influencer_id;
```

Output:

```
SELECT * FROM influencers;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25  Filter rows:

Extra options

		 influencer_id		name
☐	 Edit	 Copy	 Delete	1 abhi
☐	 Edit	 Copy	 Delete	2 velcy
☐	 Edit	 Copy	 Delete	3 harish

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

SELECT * FROM posts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Sort by

Extra options

				post_id	influencer_id	caption			
☐		Edit		Copy		Delete	1	1	My first travel vlog!
☐		Edit		Copy		Delete	2	1	Exploring a new place today!
☐		Edit		Copy		Delete	3	2	Morning workout routine!
☐		Edit		Copy		Delete	4	2	Healthy food ideas!
☐		Edit		Copy		Delete	5	3	New technology tips!
☐		Edit		Copy		Delete	6	3	Learning SQL today!

Extra options

name	caption
abhi	My first travel vlog!
abhi	Exploring a new place today!
velcy	Morning workout routine!
velcy	Healthy food ideas!
harish	New technology tips!
harish	Learning SQL today!

- 2) Write an SQL query using INNER JOIN to display each post's caption along with the name of the influencer who posted it, based on the tables you created.

Query:

```
1 SELECT
2     posts.caption,
3     influencers.name
4 FROM posts
5 INNER JOIN influencers
6 ON posts.influencer_id = influencers.influencer_id;
```

Output:

☐ Show all | Number of rows: 25 ▼

Extra options

caption	name
My first travel vlog!	abhi
Exploring a new place today!	abhi
Morning workout routine!	velcy
Healthy food ideas!	velcy
New technology tips!	harish
Learning SQL today!	harish

3) Write an SQL query using LEFT JOIN to show all influencers and their posts, including influencers who haven't posted anything yet. If an influencer has no posts, display 'No Posts' in the caption column.

Hint: Use IFNULL or COALESCE to handle NULL values in the result.

Query:

```
1 SELECT
2     influencers.name,
3     COALESCE(posts.caption, 'No Posts') AS caption
4 FROM influencers
5 LEFT JOIN posts
6 ON influencers.influencer_id = posts.influencer_id;
```

Output:

```
SELECT influencers.name, COALESCE(posts.caption, 'No P
posts.influencer_id;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)]

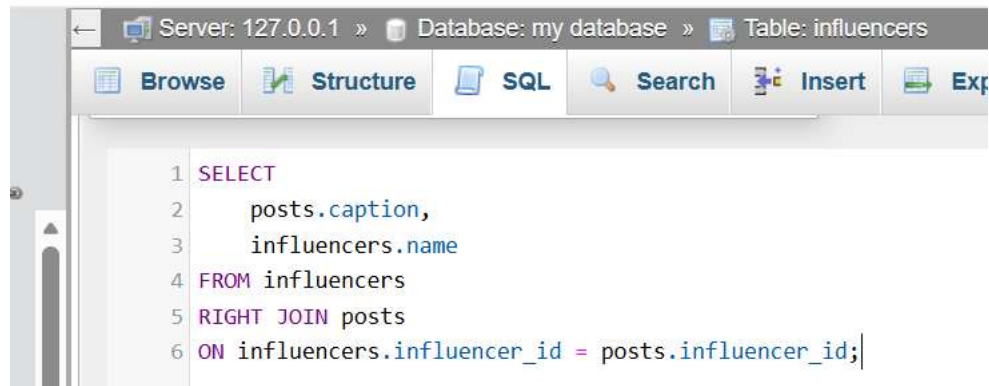
☐ Show all | Number of rows: 25 ▾ | Filter rows: \$

Extra options

name	caption
abhi	My first travel vlog!
abhi	Exploring a new place today!
velcy	Morning workout routine!
velcy	Healthy food ideas!
harish	New technology tips!
harish	Learning SQL today!

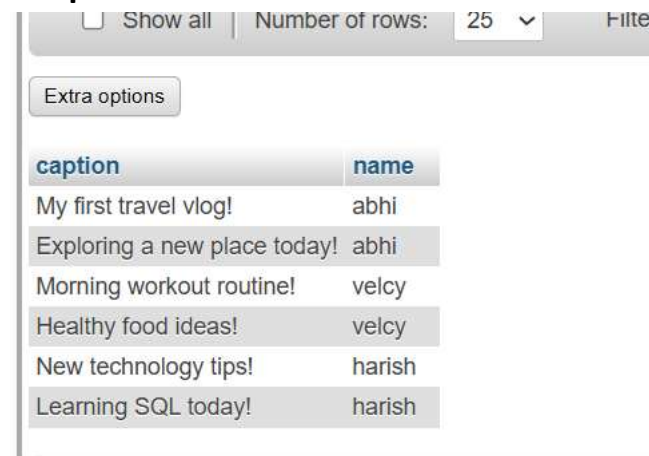
- 4) Write an SQL query using RIGHT JOIN to list all posts and the corresponding influencer's name, ensuring that even posts without a matching influencer_id (if any) are shown.

Query:



```
1 SELECT
2     posts.caption,
3     influencers.name
4 FROM influencers
5 RIGHT JOIN posts
6 ON influencers.influencer_id = posts.influencer_id;
```

Output:



Number of rows: 25

caption	name
My first travel vlog!	abhi
Exploring a new place today!	abhi
Morning workout routine!	velcy
Healthy food ideas!	velcy
New technology tips!	harish
Learning SQL today!	harish